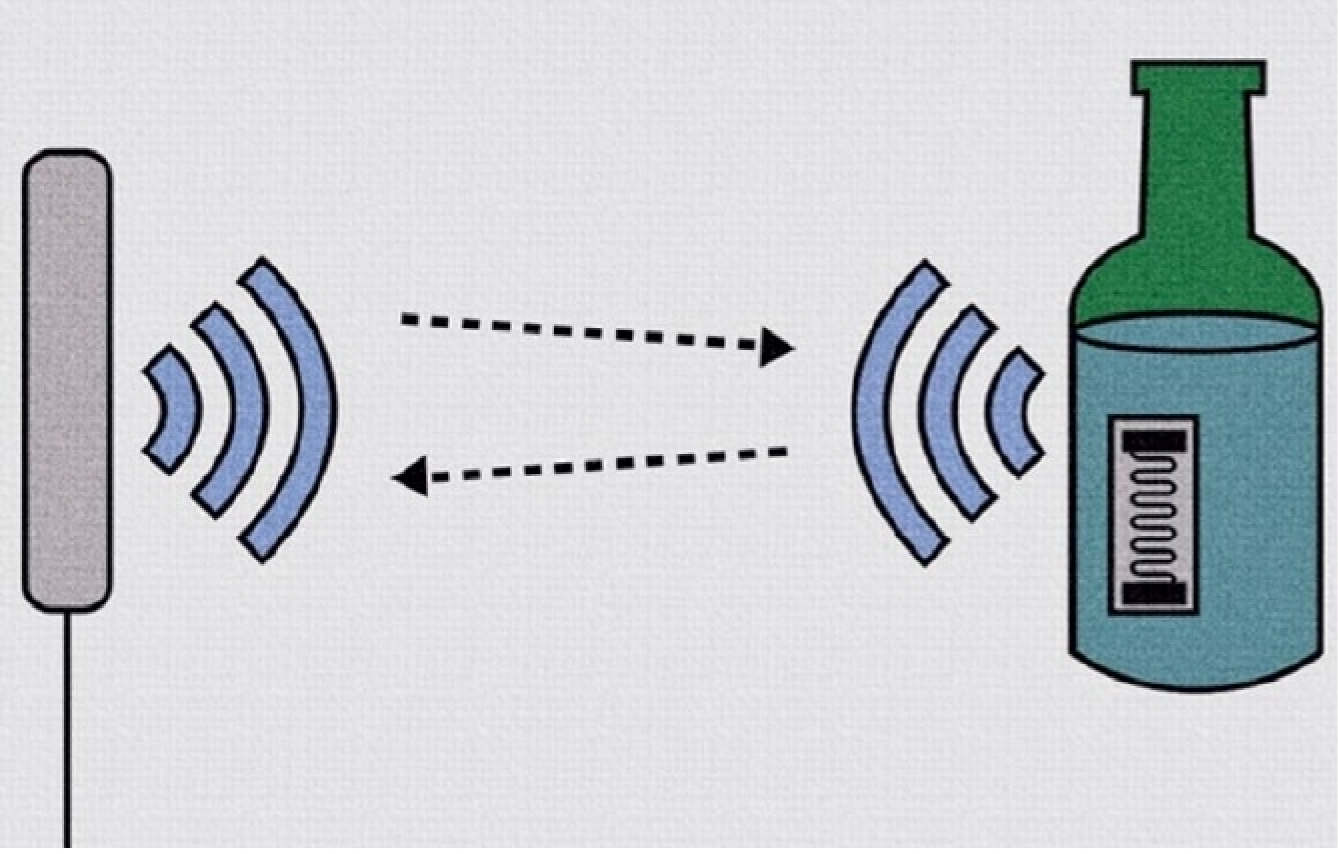
New wireless system can sense potential food contamination using cheap RFID tags

Researchers at Massachusetts Institute of Technology (MIT) Media Lab have developed a wireless system that leverages cheap RFID tags already available on billions of products to sense potential food contamination—without any hardware modifications.

The system—called RFIQ—includes a reader that senses minute changes in wireless signals emitted from RFID tags when the signals interact with food. For this study, researchers focussed on baby formula and alcohol, but in the future, consumers might have their own reader and software to conduct food safety sensing before buying virtually any product.



Using the simple, scalable system, researchers hope to bring food safety detection to general public (Credit: MIT Media Lab)

This system could also be implemented in supermarket back rooms or in smart fridges to continuously ping an RFID tag to automatically detect food spoilage.

The technology hinges on the fact that certain changes in the signals emitted from an RFID tag correspond to levels of certain contaminants within that product. A machine learning model learns those correlations and, given a new material, can predict if the material is pure or tainted, and at what concentration.

In experiments, the system detected baby formula laced with melamine with 96 per cent accuracy, and alcohol diluted with methanol with 97 per cent accuracy.

Healing kidneys with nanotechnology

Acute kidney injury (AKI)—also known as acute renal failure—produces a rapid buildup of nitrogenous wastes and decreases urine output, usually within hours or days of disease onset. Severe complications often ensue. Currently, there is no known cure for AKI.

Hao Yan, director of Biodesign Center for Molecular Design and Biomimetics and Martin D. Glick Distinguished Professor in School of Molecular Sciences at Arizona State University, and his colleagues at University of Wisconsin-Madison and in China describe a new method for treating and preventing AKI. Their technique involves the use of tiny, self-assembling forms measuring just billionths of a metre in diameter.

Their research has demonstrated that introduction of DNA origami nanostructures (DONs) protect normal

Illustration showing a diseased kidney (left) and a healthy kidney (right), after rectangular DNA nanostructures migrated and accumulated in the kidney, acting to alleviate damage due to oxidative stress (Credit: Arizona State University)